databricks

Continuous Integration (CI)

LECTURE

Executing Integration Tests with SDP and Jobs

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In this lecture, you'll learn how to execute integration tests in Databricks using **Lakeflow Spark Declarative Pipelines** expectations and Databricks Lakeflow Jobs tasks to validate data pipelines end-to-end

Executing Integration Tests

With Lakeflow Spark Declarative Pipelines (SDP) or Lakeflow Jobs

Spark Declarative Pipelines (SDP)

- Use **SDP expectations** to check pipeline's results (demo technique).

Lakeflow Jobs

- Implement it as a **Databricks Job with multiple tasks** - similarly what is typically done for non-SDP code.

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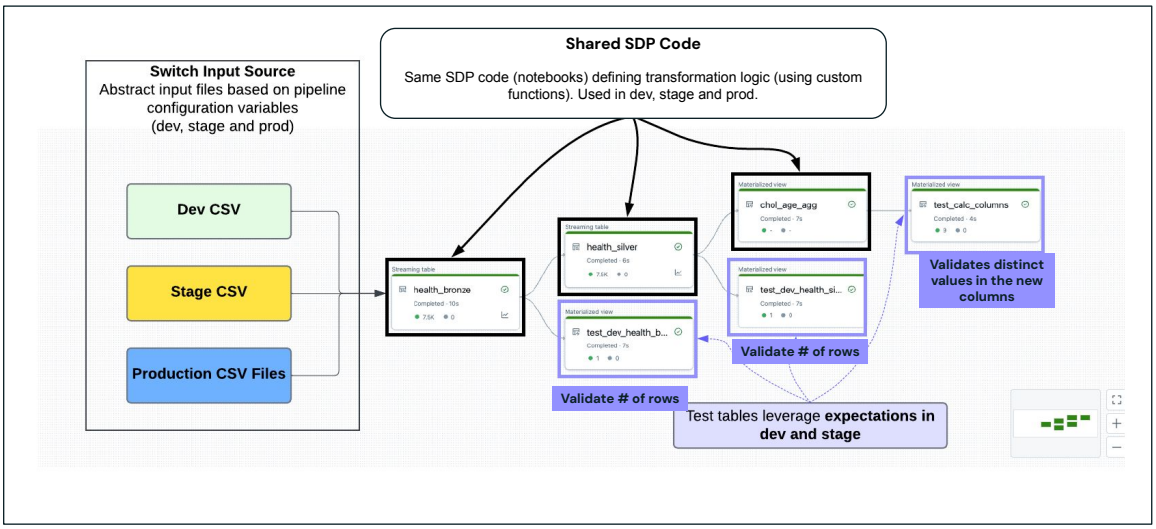
In Databricks, two simple ways to execute integration tests are with Lakeflow Spark Declarative Pipelines and Lakeflow Jobs.

With Lakeflow Spark Declarative Pipelines, we can use SDP expectations to validate the results of tables within pipelines. This is the method we will use in our project.

Alternatively, you can implement integration tests using Databricks Lakeflow Jobs by creating multiple notebooks that test specific aspects of your data pipeline. These notebooks can be added as tasks within the Lakeflow Job. While the integration tests themselves may be similar, you're using different tools in Databricks to execute them. Choose the tool that best fits your needs.

SDP – Method 1 – Expectations



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Let's example the integration test with SDP.

In this example we will use SDP to ingest CSV files from each catalog (dev, stage and prod) based on the pipeline configuration variable, it will simply read the data from the corresponding target environment.

Regardless of the target environment (dev, stage, or prod), the same SDP is executed. Focus on the black squares under the "Shared SDP Code" in the image. The same SDP transformation logic, which includes custom functions, is applied in each environment.

In this example, data from the target catalog is ingested into the "health_bronze" table, then cleaned in the "health_silver" table, and finally aggregated into a materialized view in the "chol_age_agg" gold table for each environment.

The purple boxes beneath the shared SDP represent test materialized views created with SDP expectations during the dev and stage runs. Since we're using static data for dev and stage, we know the expected output, which allows us to test our shared SDP code. In this case, we’re performing simple expectations for demonstration purposed, like counting the number of rows in the tables, to confirm correct ingestion. Typically you would create much more focused tests.

Jobs - Method 2 - Tasks

Performing unit tests to confirm all created functions work correctly.

Executing a Spark Declarative Pipeline to create the necessary tables without using expectations

Add notebooks or files within a task to perform your integration tests within a job

Unit_Tests
...for Data Engineering/Run Unit Tests

Health_ETL
Pipeline: sdk_health_etl_cruel_site...

Integration_Tests
...tabricks.com/peter_test/peter_test

Visualization
... Engineering/src/Final Visualization

Unit tests

- Testing individual units of code in isolation
- If the unit test pass continue on into the Job

Tests can include:

- Validate # of rows
- Tables were created successfully
- Columns contain correct distinct values
- Columns contain values between a specific range
- Columns contain no duplicates

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re Foundation.

Instead of using SDP and expectations for integration tests, you can also use Databricks Lakeflow Jobs with tasks.

Looking at the entire Job for this simple project, , we start by executing unit tests to test individual functions in isolation. If any unit test fails, the job will fail.

Next, we execute the SDP to ingest the data. In this example, we're running the same SDP as before but without the expectations.

After that, we perform integration tests using notebooks set as tasks within lakeflow Job. For this job, you'll need to configure the correct parameters for your target environment (dev, stage, or production). You can run a variety of integration tests, such as counting rows in a table, verifying that tables were created successfully, checking that tables contain the specified columns or distinct values, ensuring column values fall within a certain range, confirming there are no duplicates, and more.

Finally, once the unit tests, data pipeline, and integration tests are successfully executed, the final visualization is created.

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Thank you for completing this lesson and continuing your journey to develop your skills with us.